

Young Tableaux

Combinatorics

21/11/2020

1 Young Tableaux

There has been a long-existing problem in Mathematics about the number of partitions of a number n into a sum of positive integers:

Definition 1.1 Given an positive integer n , a **partition** λ of n is a sequence of positive integers $\lambda = (\lambda_1, \lambda_2, \dots, \lambda_k)$ such that $\lambda_1 \geq \lambda_2 \geq \dots \geq \lambda_k$ and $\lambda_1 + \lambda_2 + \dots + \lambda_k = n$. We also write $|\lambda| = \lambda_1 + \lambda_2 + \dots + \lambda_k = n$

Example 1.2 $\lambda = (4, 2, 2, 1)$ is a partition of 9, because $9 = 4 + 2 + 2 + 1$

Open Problem 1.3 given a positive integer n , how many partitions of n are there?

Definition 1.4 We have a method of representing a partition λ of n with a diagram, called a **Young diagram** or **Ferre diagram**, with λ_i boxes in the i -th row, aligned to the left. Note that there are two conventions: the English and the French.

For example, the Young diagram corresponding to the partition $\lambda = (4, 3, 1)$ of 8 is:



Figure 1: English convention vs French convention

Definition 1.5 We can also put in a positive integer in each box of a Young diagram with n boxes. If:

1. the entries are increasing in each row from left to right
2. the entries are strictly increasing going up column

Then we call this a **Young Tableaux**. If we further assume that the numbers in the Young Tableaux are $1, 2, \dots, n$, each used once, then we call it a **Standard Young Tableaux**. We call λ the **shape** of the (standard) Young Tableaux.

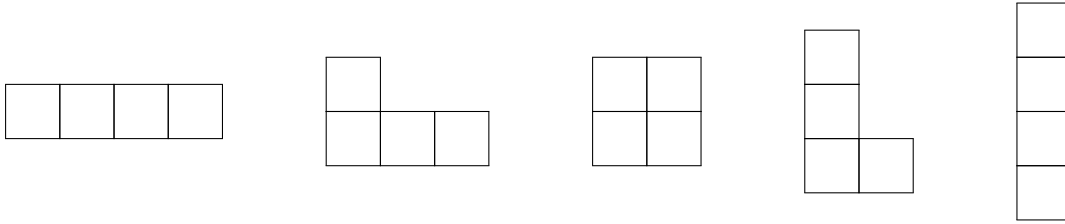
For a given partition λ , we write f^λ as the number of Standard Young Tableaux of shape λ

1	5			
1	4	4		
1	1	2	3	7

8	10			
2	4	7		
1	3	5	6	9

Figure 2: A Young Tableaux and a standard Young Tableaux

Exercise 1.6 The number 4 has partitions $4 = 3 + 1 = 2 + 2 = 2 + 1 + 1 = 1 + 1 + 1 + 1$. For each of the Young diagram, find out how many standard Young Tableaux of each shape.



Definition 1.7 For a Young diagram, the **hook** of a box (i, j) is the boxes above or on the right of itself (including itself). The hook length $h(i, j)$ is the total number of boxes in the hook of (i, j) .

2	1					
5	4	2	1			
7	6	4	3	1		
10	9	7	6	4	2	1

Figure 3: A Young diagram (not a Young Tableaux!) filled in with the hook length of each box. The hook of $(2, 3)$ is shaded, showing how $h(2, 3) = 4$

With all the preparation work done, we will hereby present our astonishing results:

Theorem 1.8 (Hook length formula) Let λ be a partition of n

$$f^\lambda = \frac{n!}{\sum_{(i,j)} h(i,j)}$$

where the sum is taken over all boxes of the Young diagram of shape λ

Theorem 1.9 For positive integer n ,

$$n! = \sum_{\lambda} (f^\lambda)^2$$

where the sum is taken over all partitions λ of n

Theorem 1.10 The total number of Young Tableaux with n boxes is:

$$\sum_{\lambda} f^\lambda = \sum_{k=0}^{\lfloor \frac{n}{2} \rfloor} \binom{n}{2k} (2k-1)!!$$

2 Prelude: Symmetry Group

Definition 2.1 A **permutation** of $1, 2, 3, \dots, n$ is a rearrangement of these n objects. (rigorously, it is a bijective function from $1, 2, 3, \dots, n$ to itself) It can be denoted by a $2 \times n$ array: for example, the array

$$\begin{pmatrix} 1 & 2 & 3 & 4 & 5 & 6 \\ 2 & 4 & 6 & 1 & 5 & 3 \end{pmatrix}$$

means that $1 \rightarrow 2 \rightarrow 4, 3 \leftrightarrow 6$ and 5 goes to itself. (it can also be written as $(124)(36)$ but we won't be using this notation here.)

Obviously, there are $n!$ permutations of $1, 2, \dots, n$. We say that the set of these permutations S_n together with the function composition $\circ$ forms a "group", which means that:

Proposition 2.2

1. There is an identity element $e = \begin{pmatrix} 1 & 2 & \dots & n \\ 1 & 2 & \dots & n \end{pmatrix}$ such that for any permutation σ , $\sigma \circ e = e \circ \sigma = \sigma$
2. For any permutations f, g, h , $(f \circ g) \circ h = f \circ (g \circ h)$
3. For any permutation σ , there is an **inverse** σ^{-1} such that $\sigma \circ \sigma^{-1} = \sigma^{-1} \circ \sigma = e$

We can calculate the compositions of permutations using the "ghost leg (劃鬼腳)" method:

Example 2.3 For the permutation $\sigma = \begin{pmatrix} 1 & 2 & 3 & 4 & 5 \\ 2 & 3 & 1 & 5 & 4 \end{pmatrix}$, $\varphi = \begin{pmatrix} 1 & 2 & 3 & 4 & 5 \\ 2 & 3 & 4 & 5 & 1 \end{pmatrix}$

- (a) Calculate $\sigma \circ \varphi$ and $\varphi \circ \sigma$, are they the same permutation?
- (b) Calculate σ^2 (defined as $\sigma \circ \sigma$)
- (c) Calculate σ^{-1}

The group *structure* we mentioned will not become very useful in this lesson, but the concept of composition and inverse will become useful in our

3 The Row Bumping Algorithm

We now define a useful algorithm, which will be useful in the next part:

Algorithm 3.1 Given a tableaux with a single row

a_1	a_2	.	.	.	a_k
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 (where $a_1 < a_2 < \dots < a_k$), and a number x , we define the algorithm of **bumping x into the row** which gives the outcome

- | | | | | | | |
|-------|-------|---|---|---|-------|-----|
| a_1 | a_2 | . | . | . | a_k | x |
|-------|-------|---|---|---|-------|-----|

 if $x > a_k$
- | | | | | | | | | | | | |
|-------|-------|---|---|---|-----------|-----|-----------|---|---|---|-------|
| a_1 | a_2 | . | . | . | a_{r-1} | x | a_{r+1} | . | . | . | a_k |
|-------|-------|---|---|---|-----------|-----|-----------|---|---|---|-------|

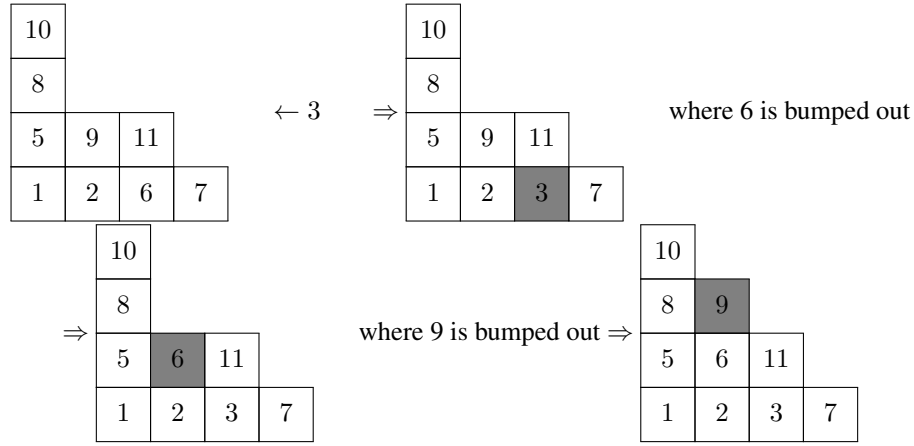
 (with the number a_r "bumped out") if a_r is the least a_i that is greater than x

Algorithm 3.2 (Row bumping algorithm) Given a standard Young Tableaux P and a number x not contained in P , we can **bump x into P** (denoted by $P \leftarrow x$), which does the followings:

- bump x into the 1st row of P
- if a number x_1 is bumped out from row 1, then bump it into row 2
- in the same way, if a number x_r is bumped out from row r , then bump it into row $r + 1$
- repeat this until no number is bumped out

sketch of proof You need to prove the validity of this algorithm: does it really turn Young Tableaux into Young Tableaux?

Example 3.3 Row bumping algorithm



Proposition 3.4 If you are given the result of $P \leftarrow x$, together with the location of the newly added box, then you can find the tableaux P back by using the following **reverse algorithm**:

- Suppose the newly added box is in the m -th row, and has number x_m written in it
- Take this number x_m from m -th row, bump it into the $(m - 1)$ -th row, but this time, we bump the *largest number less than x_m*
- Repeat this process inductively until you reach the 1st row. The number bumped away from 1st row is x we wanted, and the remaining Tableaux is P .

sketch of proof The proof is quite long, and there are quite a few things to verify:

1. The validity of this reverse algorithm
 - is it always possible for us to carry out the "bumping" in every step? (i.e. is it possible that all the numbers in row $(m - 1)$ exceed x_m ?)
 - does each bumping turns a Young Tableaux into another Young Tableaux?
2. Whether the row bumping algorithm $B : (T, x) \mapsto (T \leftarrow x)$ and the reverse algorithm $R : (T \leftarrow x) \mapsto (T, x)$ are inverses of each other
 - To do this, we only need to show that each bumping step (in the row bumping algorithm) and each inverse bumping (in the reverse algorithm) are inverse to each other.

Example 3.5 Suppose the following is the Young Tableaux $T \leftarrow x$, with the shaded box being the newly added square. Try to find T and x :

10			
6	8		
4	5	9	
1	2	3	7

4 The RSK Correspondence

Let n be an integer. The Robinson-Schensted-Knuth Correspondence is a bijective (1-to-1) correspondence between

- ordered pairs (P, Q) where P, Q are standard Young Tableaux of the same shape, each with n boxes
- permutations of $1, 2, 3, \dots, n$ (i.e. elements of S_n)

Here is how the algorithm works: given a permutation $\sigma \in S_n$, you can construct a standard Young Tableaux P by repeating the row bumping algorithm:

$$(\dots((\boxed{} \leftarrow \sigma(1)) \leftarrow \sigma(2))\dots) \leftarrow \sigma(n)$$

While Q is constructed as the *recording Tableaux*: in the $n - th$ row bumping, find the location in P where the new box appears, add a box with number n in the corresponding position in Q . (you can see the video for a better understanding of this)

Exercise 4.1 Convince yourself that P, Q constructed this way are both standard Young Tableaux

The inverse algorithm is quite easy to define: just repeatedly perform the reverse algorithm to P (this is possible because Q helps you keep track of when and where new boxes appear)

Although this correspondence seemed random and unmotivated at the first glance, it actually has some beautiful properties! Here is one of them:

Theorem 4.2 (Symmetry Theorem) If (P, Q) correspond to the permutation σ , then (Q, P) correspond to the permutation σ^{-1}

sketch of proof (as in the video)

- Construct a new RSK correspondence using the light-and-shadow algorithm
- (sidetrack) You can recover the original permutation (black points) by looking at the "skeleton" (red points) of the permutation
- (sidetrack) A **Yamanouchi word** is a word formed with numbers $1, 2, \dots, n$ such that for any $i < j$, and for any k , among the first k letters of this word, the number i appears more or equal times than the number j . There is a natural way that correspond a Yamanouchi word of length n with a standard Young Tableaux with n boxes
- Under this new RSK correspondence, the symmetry theorem become obvious (because everything is just reflection along the diagonal)
- Prove that this new RSK correspondence is the same as the old RSK correspondence

5 Reaping Time!

With our main correspondence proved, we can now prove the results we mentioned in the beginning of lesson much more easily:

Exercise 5.1 It might be better to have a hands on experience of RSK algorithm before you really start

- (a) Under the RSK algorithm, what (P, Q) does the permutation $\begin{pmatrix} 1 & 2 & 3 & 4 & 5 & 6 \\ 2 & 4 & 1 & 5 & 6 & 3 \end{pmatrix}$ correspond to?
- (b) Under the RSK algorithm, what permutation does the pair of standard Young Tableaux

5				3			
4	6			2	6		
1	2	3		1	4	5	

correspond to?

Exercise 5.2 Show that For positive integer n ,

$$n! = \sum_{\lambda} (f^{\lambda})^2$$

Exercise 5.3 The total number of Young Tableaux with n boxes is:

$$\sum_{\lambda} f^{\lambda} = \sum_{k=0}^{\lfloor \frac{n}{2} \rfloor} \binom{n}{2k} (2k-1)!!$$

Hint: the following definition might be useful:

Definition 5.3.1 A permutation $\sigma \in S_n$ of $1, 2, \dots, n$ is called an **involution** if $\sigma^2 = 1$

Exercise 5.4 (hook length formula)

- (a) Let $\lambda = (\lambda_1, \lambda_2, \dots, \lambda_k)$, and we let $l_i = \lambda_i + k - i$ (then $l_1 > l_2 > \dots > l_k$)
Show that the hook length formula (Theorem 1.8) is equivalent to the following:

$$f^{\lambda} = \frac{n! \cdot \prod_{i < j} (l_i - l_j)}{l_1! \cdot l_2! \cdot \dots \cdot l_k!}$$

- (b) Show that

$$f^{\lambda} = \sum_{i=1}^k f^{(\lambda_1, \lambda_2, \dots, \lambda_i - 1, \dots, \lambda_k)}$$

- (c) Complete the proof of Hook Length Formula with strong induction.